Market Basket Analysis

Identifying Prescription Relationships in Hospital Patient Data

Research Question

Is it possible to analyze hospital medical data using market basket analysis to determine if there are relevant relationships between prescriptions doctors could use to more quickly and effectively treat patients?

Analysis Goal

This data analysis aims to uncover at least one relationship between widely used prescriptions that can be clearly explained to doctors.

Method Justification

Market basket analysis will analyze this dataset by finding relationships between the occurrences of the prescriptions patients require in relation to one another. MBA will be searching for a relationship in the likelihood of the presence of one prescription based on the known occurrence of another. MBA will also compare the 'expected' rate of prescriptions to the actual rate found in the data, analyzing whether prescription combinations are occurring more or less often than would be expected, which can reveal relationships between medications.

The expected outcome of this analysis is that the strongest relationships between services/medications will be determined and numerically defined.

Market basket analysis assumes that items are purchased independently of one another.

Example Transaction

In this dataset, a transaction will be all the prescriptions of a patient, which will be one row of the initial medical dataset. For example, this row containing citalopram, Benicar, and amphetamine salt combo XR is one 'transaction'.

Presc01	Presc02	Presc03	Presc04
amlodipine	albuterol aerosol	allopurinol	pantoprazole
citalopram	benicar	amphetamine salt combo xr	

Apriori Algorithm Output

```
    ...:
    ...: rules = apriori(data, min_support = 0.02, use_colnames=True)
    ...: print(rules.head(5))
    support              itemsets
0  0.046794            (Premarin)
1  0.238368             (abilify)
2  0.020397  (albuterol aerosol)
3  0.033329         (allopurinol)
4  0.079323          (alprazolam)

In [13]:
    ...: rul_table = association_rules(rules, metric='lift', min_threshold=1)
    ...: print(rul_table.head(20))
                   antecedents  ... kulczynski
0                    (abilify)  ...   0.214609
1                 (amlodipine)  ...   0.214609
2                    (abilify)  ...   0.229537
3      (amphetamine salt combo)  ...   0.229537
4                    (abilify)  ...   0.248515
5   (amphetamine salt combo xr)  ...   0.248515
6                    (abilify)  ...   0.285856
7               (atorvastatin)  ...   0.285856
8                    (abilify)  ...   0.296796
9                 (carvedilol)  ...   0.296796
10                   (abilify)  ...   0.207130
11                    (cialis)  ...   0.207130
12                   (abilify)  ...   0.191083
13                (citalopram)  ...   0.191083
14                   (abilify)  ...   0.237819
15               (clopidogrel)  ...   0.237819
16                   (abilify)  ...   0.227014
17       (dextroamphetamine XR)  ...   0.227014
18                   (abilify)  ...   0.271158
19                  (diazepam)  ...   0.271158
[20 rows x 14 columns]

In [14]:
    ...: rul_table.to_csv("C:/Users/velez/OneDrive/Desktop/School/D212/Task 3/
rul_table.csv",index=False)

In [15]:
    ...: top_three_rules_conf = rul_table.sort_values('confidence', ascending=False).head(3)
    ...: print(top_three_rules_conf)
     antecedents consequents  ...  certainty  kulczynski
31   (metformin)   (abilify)  ...   0.286354    0.276610
25   (glipizide)   (abilify)  ...   0.237201    0.267400
29  (lisinopril)   (abilify)  ...   0.233952    0.294127

[3 rows x 14 columns]

In [16]:
    ...: top_three_rules_conf.to_csv("C:/Users/velez/OneDrive/Desktop/School/D212/Task 3/
top_three_rules_conf.csv",index=False)

In [17]:
    ...: top_three_rules_lift = rul_table.sort_values('lift', ascending=False).head(3)
    ...: print(top_three_rules_lift)
    ...:
    ...: top_three_rules_lift.to_csv("C:/Users/velez/OneDrive/Desktop/School/D212/Task 3/
top_three_rules_lift.csv",index=False)
     antecedents   consequents  ...  certainty  kulczynski
74  (carvedilol)  (lisinopril)  ...   0.140684    0.312015
75  (lisinopril)  (carvedilol)  ...   0.272197    0.312015
73   (glipizide)  (carvedilol)  ...   0.210764    0.239939

[3 rows x 14 columns]
```

```
[3 rows x 14 columns]

In [16]:
    ...: top_three_rules_conf.to_csv("C:/Users/velez/OneDrive/Desktop/School/D212/Task 3/
top_three_rules_conf.csv",index=False)

In [17]:
    ...: top_three_rules_lift = rul_table.sort_values('lift', ascending=False).head(3)
    ...: print(top_three_rules_lift)
    ...:
    ...: top_three_rules_lift.to_csv("C:/Users/velez/OneDrive/Desktop/School/D212/Task 3/
top_three_rules_lift.csv",index=False)
     antecedents   consequents  ...  certainty  kulczynski
74  (carvedilol)  (lisinopril)  ...   0.140684    0.312015
75  (lisinopril)  (carvedilol)  ...   0.272197    0.312015
73   (glipizide)  (carvedilol)  ...   0.210764    0.239939

[3 rows x 14 columns]

In [18]:
    ...: top_three_rules_supp = rul_table.sort_values('support', ascending=False).head(3)
    ...: print(top_three_rules_supp)
    ...:
    ...: top_three_rules_supp.to_csv("C:/Users/velez/OneDrive/Desktop/School/D212/Task 3/
top_three_rules_Supp.csv",index=False)
     antecedents   consequents  ...  certainty  kulczynski
8      (abilify)  (carvedilol)  ...   0.092566    0.296796
9   (carvedilol)     (abilify)  ...   0.137421    0.296796
19    (diazepam)     (abilify)  ...   0.109018    0.271158

[3 rows x 14 columns]

In [19]:
```

Top Three Association Rules

antecedents	consequents	antecedent support	consequent support	support	confidence	lift	representativity	leverage	conviction	zhangs_metric	jaccard	certainty
frozenset({'amlodipine'})	frozenset({'abilify'})	0.071457139	0.238368218	0.023596854	0.330223881	1.385351973	1	0.006563743	1.137143831	0.299568058	0.082440615	0.12060
frozenset({'abilify'})	frozenset({'amlodipine'})	0.238368218	0.071457139	0.023596854	0.098993289	1.385351973	1	0.006563743	1.030561537	0.365218191	0.082440615	0.02965
frozenset({'amphetamine salt combo'})	frozenset({'abilify'})	0.068390881	0.238368218	0.024396747	0.356725146	1.496529822	1	0.008094535	1.183991225	0.356144491	0.086402266	0.15539

The first rule generated by the Apriori algorithm is between 'amlodipine' and 'abilify'. This rule has a support value of 0.0235, meaning this combination is found in 2.35% of patients. This relationship has a confidence value of 0.3302, meaning 33.02% of patients who are prescribed amlodipine are also prescribed abilify. The lift score of 1.38 is indicative that patients are on a combination of these medications 1.38 times more often than what we would have expected, indicating some relationship exists between the two medications.

The second rule generated by the Apriori algorithm is between 'abilify' and 'amlodipine'. This rule has a support value of 0.0235, meaning this combination is found in 2.35% of patients. This relationship has a confidence value of 0.0989, meaning 9.89% of patients who are prescribed abilify are also prescribed amlodipine. The lift score of 1.38 is indicative that patients are on a combination of these medications 1.38 times more often than what we would have expected, indicating some relationship exists between the two medications.

The third rule generated by the Apriori algorithm is between 'amphetamine salt combo' and 'abilify'. This rule has a support value of 0.0243, meaning this combination is found in 2.43% of patients. This relationship has a confidence value of 0.3567, meaning 35.67% of patients who are prescribed amphetamine salt combo are also prescribed abilify. The lift score of 1.49 is indicative that patients are on a combination of these medications 1.49 times more often than what we would have expected, indicating some relationship exists between the two medications.

Support, Lift, and Confidence

```
   ...:
   ...: rules = apriori(data, min_support = 0.02, use_colnames=True)
   ...: print(rules.head(5))
 support              itemsets
0.046794           (Premarin)
0.238368            (abilify)
0.020397  (albuterol aerosol)
0.033329        (allopurinol)
0.079323         (alprazolam)
```

Support refers to how many instances occur in the dataset. The support value for an individual prescription, such as 0.046 for Premarin indicates that 4.6% of patients in this dataset require this medication. Combinations of medications will also have a support value, such as the combo Abilify and Carvedilol resulting in support value of 0.059, indicating 5.9% of patients are prescribed both medications simultaneously.

A	B	C	D	E	F	
antecedents	consequents	antecedent support	consequent support	support	confidence	li
frozenset({'abilify'})	frozenset({'carvedilol'}	0.238368218	0.174110119	0.05972537	0.250559284	
frozenset({'carvedilol'}	frozenset({'abilify'})	0.174110119	0.238368218	0.05972537	0.343032159	
frozenset({'abilify'})	frozenset({'diazepam'}	0.238368218	0.163844821	0.052659645	0.220917226	

Confidence refers to the boost in likelihood of a second prescription based on a known prescription. The confidence value of 0.456 of Metformin to Abilify shows that there is a 45.6% increase in likelihood of a patient receiving an abilify prescription if prescribed metformin.

antecedents	consequents	antecedent support	consequent support	support	confidence	lift
frozenset({'metformin'})	frozenset({'abilify'})	0.050526596	0.238368218	0.023063592	0.45646438	1.914954874
frozenset({'glipizide'})	frozenset({'abilify'})	0.065857886	0.238368218	0.02759632	0.41902834	1.757903568

antecedents	consequents	antecedent support	consequent support	support	confidence	lift	repre
frozenset({'carvedilol'})	frozenset({'lisinopril'})	0.174110119	0.098253566	0.039194774	0.225114855	2.291162176	
frozenset({'lisinopril'})	frozenset({'carvedilol'})	0.098253566	0.174110119	0.039194774	0.398914518	2.291162176	
frozenset({'glipizide'})	frozenset({'carvedilol'})	0.065857886	0.174110119	0.022930276	0.348178138	1.999758201	

Lift shows the comparison between the actual rate of two prescriptions occurring simultaneously and the expected rate of this occurrence. This is done by dividing the actual rate by the expected rate and comparing this value to 1. A value greater than 1 indicates a relationship between the two, having one prescription would increase the likelihood of another. The lift value of 2.29 between Carvedilol and lisinopril shows that the prescription of both medications occurs 2.29 more likely than the expected rate, indicating a relevant relationship is at play.

Practical Significance

This analysis has uncovered many relationships between prescriptions, such as strong lift values for combinations of carvedilol and lisinopril, high confidence values for the relationship between metformin and abilify, and base level understandings of how common each medication or medication combination is by providing each with support values.

Course of Action

It is recommended that based on the findings from this analysis that doctors should be made aware of the strongest relationships between medications in order to get a step ahead, if 45.6% of patients being prescribed Metformin are later prescribed Abilify, then doctors should know that upon prescribing Metformin, they should also be looking out for potential benefits of adding in Abilify that many other patients have successfully relied on. In the same way, doctors should be made aware of high lift values for relationships among medications, potentially prescribing both upfront when necessary. The lift value of 2.29 between carvedilol and lisinopril should give doctors some context that many patients will need both and spotlight a potential combination solution that many others have used.